<Intro to Machine Learning- Problem Set 1>

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1. R/Python and exploratory data analysis: Chapter 2: #10

a) How many rows are in this data set? How many columns? What do the rows and columns represent?

(code)

```
library(ISLR2)
Boston
?Boston
```

(comment)

Rows : 506, Columns: 13

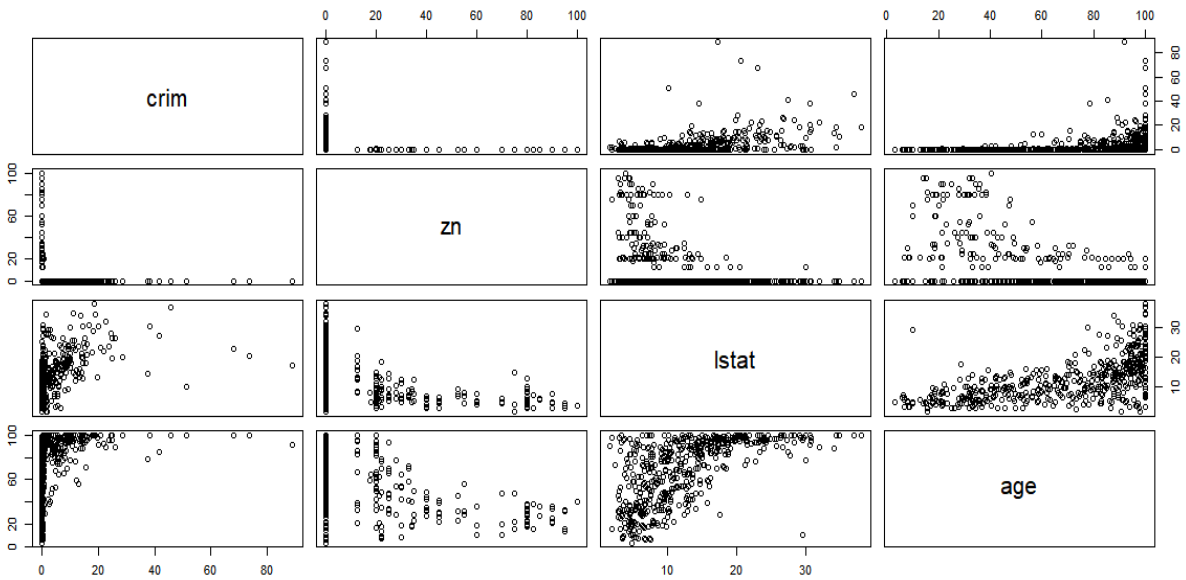
Each row represents a different suburb in the Boston area,

Each column represents different variables describing characters of that suburb such as crime rate, average number of rooms, property tax rate

b) Make some pairwise scatterplots of the predictors (columns) in this data set. Describe your findings.

(code)

```
pairs(Boston[,c('crim','zn','lstat','age')])
```



(comment)

- crim is positively correlated with lstat and age, implicating that suburbs with older housing and higher low-income populations tend to have higher crime rates.
- zn shows a negative relationship with crim and lstat, implicating that more larger zoning is associated with lower crime and low-income percentages.
- lstat and age are positively related, indicating that older neighborhoods may have higher proportions of low-income residents.

c) Are any of the predictors associated with per capita crime rate? If so, explain the relationship.

(comment)

- Yes, some predictors are correlated with crime rate(crim)

  "lstat" is positively correlated with crim, indicating that suburbs with more low-income people tend to have higher crime rates.

  "age" also shows a positive trend with crim

  "zn" is negatively correlated with crim, suggesting that more affluent areas experience less crime.

d) Do any of the census tracts of Boston appear to have particularly high crime rates? Tax rates? Pupil-teacher ratios? Comment on the range of each predictor.

(code)

```
range(Boston$crim)
range(Boston$tax)
range(Boston$ptratio)
which.max(Boston$crim)
which.max(Boston$tax)
which.max(Boston$ptratio)
```

(comment)

The highest crime rate is in tract 381 with crim=88.98

(crim range : 0.006 to 88.98)

The highest tax rate is in tract 488 with tax=711

(tax range : 187 to 711)

The highest pupil-teacher ratio is in tract 355 with ptratio=22.0

(tax range : 12.6 to 22.0)

e) How many of the census tracts in this data set bound the Charles river?

(code)

```
sum(Boston$chas==1)
```

(comment)

There are 35 census tracts in the Boston data set that bound the Charles River.

f) What is the median pupil-teacher ratio among the towns in this data set?

(code)

```
median(Boston$ptratio)
```

(comment)

The median pupil-teacher ratio among the towns in Boston data set is 19.05

g) Which census tract of Boston has lowest median value of owneroccupied homes? What are the values of the other predictors for that census tract, and how do those values compare to the overall ranges for those predictors? Comment on your findings.

(code)

```
which.min(Boston$medv)
Boston[which.min(Boston$medv),]
```

(comment)

The census tract with the lowest median value of owner-occupied homes is tract 399, where medv=5.0. The zn value for this tract is 0, indicating that none of the land in this tract is zoned for large residential lots. The age value is 100, meaning 100% of the homes in this are very old. This tract has the highest value (24) in rad, indicating strong proximity to major roads.

h) In this data set, how many of the census tracts average more than seven rooms per dwelling? More than eight rooms per dwelling? Comment on the census tracts that average more than eight rooms per dwelling.

(code)

```
sum(Boston$rm>7)
sum(Boston$rm>8)
Boston[Boston$rm>8,]
range(Boston$lstat)
```

(comment)

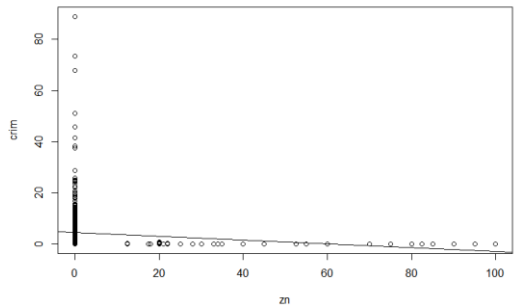
64 census tracts have an average of more than 7 rooms per dwelling.
13 census tracts have an average of more than 8 rooms per dwelling.
The tracts with more than 8 rooms per dwelling likely represent affluent neighborhoods with larger homes. These tracts tend to have higher medv, lower crim and lower lstat.
Thus, those areas are generally more expensive and have better living conditions.

2. Linear Models: Chapter 3: #15

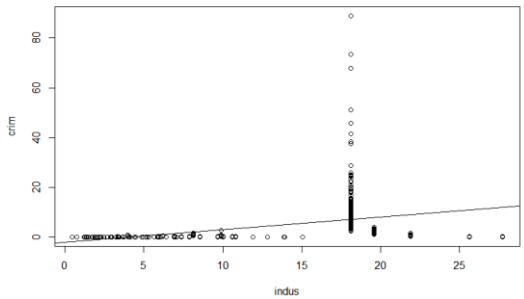
a) For each predictor, ft a simple linear regression model to predict the response. Describe your results. In which of the models is there a statistically significant association between the predictor and the response? Create some plots to back up your assertions.

(code)

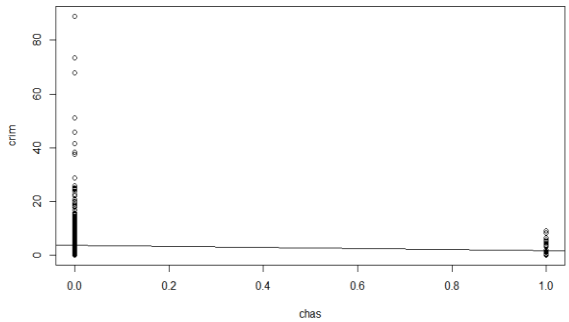
```
model_zn<-lm(crim~zn, data=Boston)
summary(model_zn)
plot(crim~zn, data=Boston)
abline(lm(crim~zn, data=Boston))
```



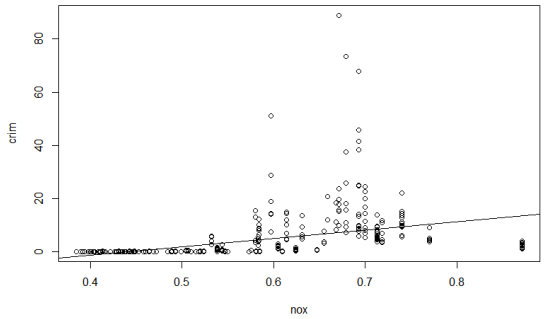

```
model_indus<-lm(crim~indus,data=Boston)
summary(model_indus)
plot(crim~indus, data=Boston)
abline(lm(crim~indus, data=Boston))
```



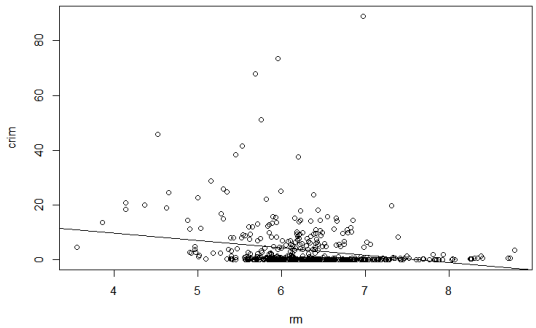

```
model_chas<-lm(crim~chas,data=Boston)
summary(model_chas)
plot(crim~chas, data=Boston)
abline(lm(crim~chas, data=Boston))
```



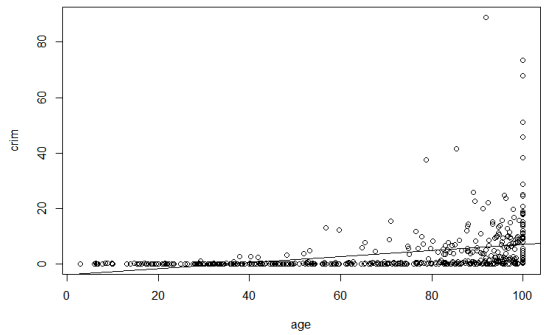

```
model_nox<-lm(crim~nox, data=Boston)
summary(model_nox)
plot(crim~nox, data=Boston)
abline(lm(crim~nox, data=Boston))
```



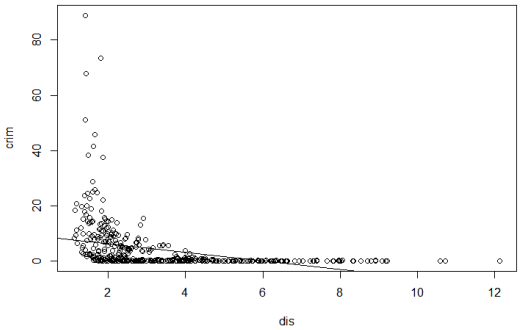

```
model_rm<-lm(crim~rm, data=Boston)
summary(model_rm)
plot(crim~rm,data=Boston)
abline(lm(crim~rm,data=Boston))
```



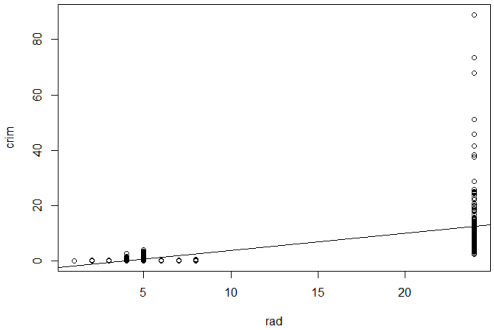

```
model_age<-lm(crim~age,data=Boston)
summary(model_age)
plot(crim~age,data=Boston)
abline(lm(crim~age,data=Boston))
```



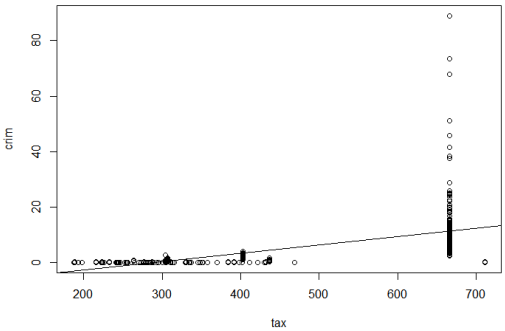

```
model_dis<-lm(crim~dis,data=Boston)
summary(model_dis)
plot(crim~dis, data=Boston)
abline(lm(crim~dis, data=Boston))
```



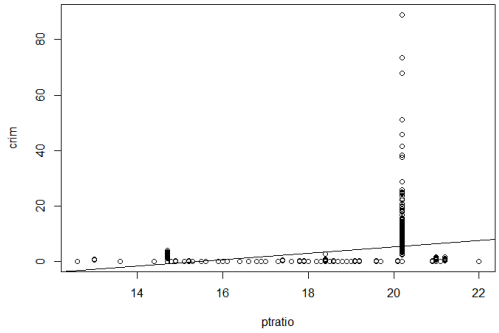

```
model_rad<-lm(crim~rad,data=Boston)
summary(model_rad)
plot(crim~rad,data=Boston)
abline(lm(crim~rad,data=Boston))
```



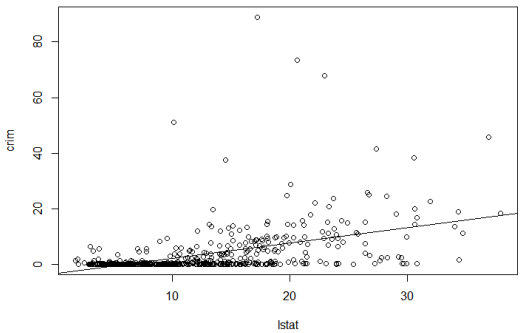

```
model_tax<-lm(crim~tax,data=Boston)
summary(model_tax)
plot(crim~tax,data=Boston)
abline(lm(crim~tax,data=Boston))
```



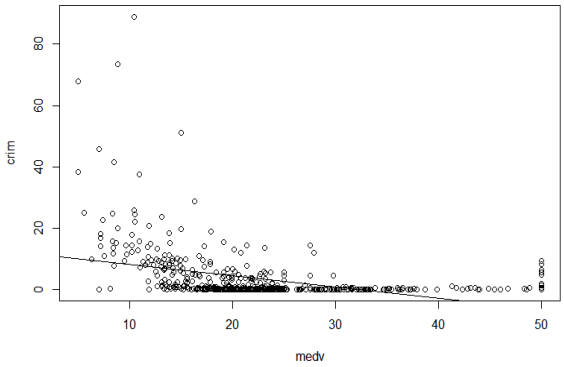

```
model_ptratio<-lm(crim~ptratio,data=Boston)
summary(model_ptratio)
plot(crim~ptratio,data=Boston)
abline(lm(crim~ptratio,data=Boston))
```




```
model_lstat<-lm(crim~lstat,data=Boston)
summary(model_lstat)
plot(crim~lstat,data=Boston)
abline(lm(crim~lstat,data=Boston))
```




```
model_medv<-lm(crim~medv,data=Boston)
summary(model_medv)
plot(crim~medv,data=Boston)
abline(lm(crim~medv,data=Boston))
```



(comment)

- There is a statistically significant negative relationship between zn and crim ($p < 0.001$).
- There is a positive and statistically significant relationship between indus and crim ($p<0.001$).
- There is a statistically insignificant negative relationship between chas and crim ($p = 0.209$).
- There is a statistically significant positive relationship between nox and crim ($p < 0.001$).
- There is a statistically significant negative relationship between rm and crim ($p < 0.001$).
- There is statistically significant positive relationship between age and crim ($p<0.001$).
- There is a statistically significant negative relationship between crim and dis($p<0.001$).
- There is a statistically significant positive relationship between rad and crim ($p < 0.001$),

but the linear relationship is weak due to a cluster of points above 20.

- There is a statistically significant positive relationship between tax and crim ($p < 0.001$), but the linear relationship is weak due to a cluster of points between 600 and 700.
- There is a statistically significant positive relationship between ptratio and crim ($p < 0.001$), but the linear relationship is weak due to a cluster of points near 14 and 20.
- There is a statistically significant positive relationship between lstat and crim ($p < 0.001$).
- There is a statistically significant negative relationship between medv and crim ($p < 0.001$).

b) Fit a multiple regression model to predict the response using all of the predictors. Describe your results. For which predictors can we reject the null hypothesis H0 : βj = 0?

(code)

```
model_all=lm(crim~.,data=Boston)
summary(model_all)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  17.033228   7.234903   2.354 0.018949 *
zn            0.044855   0.018734   2.394 0.017025 *
indus        -0.063855   0.083407  -0.766 0.444294
chas         -0.749134   1.180147  -0.635 0.525867
nox         -10.313535   5.275536  -1.955 0.051152 .
rm            0.430131   0.612830   0.702 0.483089
age           0.001452   0.017925   0.081 0.935488
dis          -0.987176   0.281817  -3.503 0.000502 ***
rad           0.588209   0.088049   6.680 6.46e-11 ***
tax          -0.003780   0.005156  -0.733 0.463793
ptratio      -0.271081   0.186450  -1.454 0.146611
black        -0.007538   0.003673  -2.052 0.040702 *
lstat         0.126211   0.075725   1.667 0.096208 .
medv         -0.198887   0.060516  -3.287 0.001087 **
```

(comment)

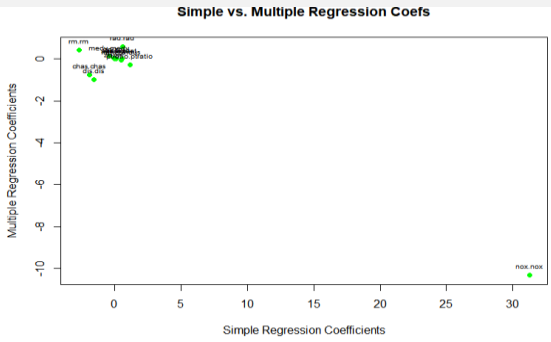
We can reject null hypothesis for zn,dis,rad and medv at the 0.05 significance level. These predictors have a statistically important influence on crim.

c) How do your results from (a) compare to your results from (b)? Create a plot displaying the univariate regression coeffcients from (a) on the x-axis, and the multiple regression coeffcients from (b) on the y-axis. That is, each predictor is displayed as a single point in the plot. Its coeffcient in a simple linear regression model is shown on the x-axis, and its coeffcient estimate in the multiple linear regression model is shown on the y-axis.

(code)

```
simple_coefs<-c( zn=coef(model_zn)[2], indus=coef(model_indus)[2], chas=coef(model_chas)[2],
nox=coef(model_nox)[2], rm=coef(model_rm)[2], age=coef(model_age)[2],
dis=coef(model_dis)[2], rad=coef(model_rad)[2], tax=coef(model_tax)[2],
ptratio=coef(model_ptratio)[2], lstat=coef(model_lstat)[2], medv=coef(model_medv)[2])

model_all=lm(crim~.,data=Boston)
multi_coefs<-coef(model_all)[-1]

valid<-is.finite(simple_coefs)&is.finite(multi_coefs)

plot(simple_coefs[valid],multi_coefs[valid], xlab ="Simple Regression Coefficients",
     ylab="Multiple Regression Coefficients",
     main="Simple vs. Multiple Regression Coefs", pch=19, col="green")
```

```
text(simple_coefs, multi_coefs, labels = names(simple_coefs), pos = 3, cex = 0.6)
```



(comment)

In the simple regressions (a), each variable was considered alone, so some effects were likely overestimated. In the multiple regression (b), all variables were considered together, revealing more realistic and adjusted effects. For example, nox showed a strong positive relationship in simple regression but switched to a negative effect in the multiple regression.

d) Is there evidence of non-linear association between any of the predictors and the response?

To answer this question, for each predictor X, ft a model of the form

$Y = \beta 0 + \beta 1X + \beta 2X2 + \beta 3X3 + \epsilon$.

(code)

```
model_age_poly<-lm(crim~age+I(age^2)+I(age^3), data=Boston)
summary(model_age_poly)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.549e+00  2.769e+00  -0.920  0.35780
age          2.737e-01  1.864e-01   1.468  0.14266
I(age^2)    -7.230e-03  3.637e-03  -1.988  0.04738 *
I(age^3)     5.745e-05  2.109e-05   2.724  0.00668 **
```

```
model_zn_poly<-lm(crim~zn+I(zn^2)+I(zn^3), data=Boston)
summary(model_zn_poly)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.846e+00  4.330e-01  11.192  < 2e-16 ***
zn          -3.322e-01  1.098e-01  -3.025  0.00261 **
I(zn^2)      6.483e-03  3.861e-03   1.679  0.09375 .
I(zn^3)     -3.776e-05  3.139e-05  -1.203  0.22954
```

```
model_indus_poly<-lm(crim~indus+I(indus^2)+I(indus^3), data=Boston)
summary(model_indus_poly)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.6625683  1.5739833   2.327   0.0204 *
indus       -1.9652129  0.4819901  -4.077 5.30e-05 ***
I(indus^2)   0.2519373  0.0393221   6.407 3.42e-10 ***
I(indus^3)  -0.0069760  0.0009567  -7.292 1.20e-12 ***
```

```
model_chas_poly<-lm(crim~chas+I(chas^2)+I(chas^3), data=Boston)
summary(model_chas_poly)
```

```
Coefficients: (2 not defined because of singularities)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.7444     0.3961   9.453   <2e-16 ***
chas         -1.8928     1.5061  -1.257    0.209
I(chas^2)         NA         NA      NA       NA
I(chas^3)         NA         NA      NA       NA
```

```
model_nox_poly<-lm(crim~nox+I(nox^2)+I(nox^3), data=Boston)
summary(model_nox_poly)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   233.09      33.64   6.928 1.31e-11 ***
nox         -1279.37     170.40  -7.508 2.76e-13 ***
I(nox^2)     2248.54     279.90   8.033 6.81e-15 ***
I(nox^3)    -1245.70     149.28  -8.345 6.96e-16 ***
```

```
model_rm_poly<-lm(crim~rm+I(rm^2)+I(rm^3), data=Boston)
summary(model_rm_poly)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 112.6246    64.5172   1.746   0.0815 .
rm          -39.1501    31.3115  -1.250   0.2118
I(rm^2)       4.5509     5.0099   0.908   0.3641
I(rm^3)      -0.1745     0.2637  -0.662   0.5086
```

```
model_dis_poly<-lm(crim~dis+I(dis^2)+I(dis^3), data=Boston)
summary(model_dis_poly)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  30.0476     2.4459  12.285  < 2e-16 ***
dis         -15.5543     1.7360  -8.960  < 2e-16 ***
I(dis^2)      2.4521     0.3464   7.078 4.94e-12 ***
I(dis^3)     -0.1186     0.0204  -5.814 1.09e-08 ***
```

```
model_rad_poly<-lm(crim~rad+I(rad^2)+I(rad^3), data=Boston)
summary(model_rad_poly)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.605545   2.050108  -0.295    0.768
rad          0.512736   1.043597   0.491    0.623
I(rad^2)    -0.075177   0.148543  -0.506    0.613
I(rad^3)     0.003209   0.004564   0.703    0.482
```

```
model_tax_poly<-lm(crim~tax+I(tax^2)+I(tax^3), data=Boston)
summary(model_tax_poly)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.918e+01  1.180e+01   1.626    0.105
tax         -1.533e-01  9.568e-02  -1.602    0.110
I(tax^2)     3.608e-04  2.425e-04   1.488    0.137
I(tax^3)    -2.204e-07  1.889e-07  -1.167    0.244
```

```
model_ptratio_poly<-lm(crim~ptratio+I(ptratio^2)+I(ptratio^3), data=Boston)
summary(model_ptratio_poly)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  477.18405  156.79498   3.043  0.00246 **
ptratio      -82.36054   27.64394  -2.979  0.00303 **
I(ptratio^2)   4.63535    1.60832   2.882  0.00412 **
I(ptratio^3)  -0.08476    0.03090  -2.743  0.00630 **
```

```
model_lstat_poly<-lm(crim~lstat+I(lstat^2)+I(lstat^3), data=Boston)
summary(model_lstat_poly)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.2009656  2.0286452   0.592   0.5541
lstat       -0.4490656  0.4648911  -0.966   0.3345
I(lstat^2)   0.0557794  0.0301156   1.852   0.0646 .
I(lstat^3)  -0.0008574  0.0005652  -1.517   0.1299
```

```
model_medv_poly<-lm(crim~medv+I(medv^2)+I(medv^3), data=Boston)
summary(model_medv_poly)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 53.1655381  3.3563105  15.840  < 2e-16 ***
medv        -5.0948305  0.4338321 -11.744  < 2e-16 ***
I(medv^2)    0.1554965  0.0171904   9.046  < 2e-16 ***
I(medv^3)   -0.0014901  0.0002038  -7.312 1.05e-12 ***
```

(comment)

- There is evidence of a non-linear association between age and crim.
  The squared and cubic terms of age are statistically significant ($p < 0.05$), indicating a curved relationship rather than a simple linear one.
- Only the linear term is significant ($p = 0.0026$), so there's no strong evidence of a nonlinear relationship between zn and crim.
- There is strong evidence of a nonlinear relationship between indus and crim — all terms including the squared and cubic terms are statistically significant ($p < 0.001$).
- chas is binary, so chas^2 and chas^3 are redundant. No non-linear relationship can be tested.
- There is strong evidence of a non-linear relationship between nox and crim (all polynomial terms are highly significant, $p < 0.001$).
- There is no strong evidence of a non-linear relationship between rm and crim, as none of the polynomial terms are statistically significant (all p-values > 0.2)
- There is strong evidence of a non-linear relationship between dis and crim, as the squared and cubic terms are both highly statistically significant ($p < 0.001$).
- There is no statistically significant linear or non-linear relationship between rad and crim, as all p-values are greater than 0.05.
- There is no statistically significant linear or non-linear relationship between tax and crim, as all p-values are greater than 0.05.
- There is evidence of a statistically significant non-linear relationship between ptratio and crim, as all polynomial terms (linear, squared, and cubed) have p-values < 0.01.
- There is no statistically significant linear or non-linear relationship between lstat and crim (all $p > 0.05$).
- There is strong statistical evidence of a non-linear relationship between medv and crim (all $p < 0.001$).

3. Shrinkage and selection in linear models: Chapter 6: #11

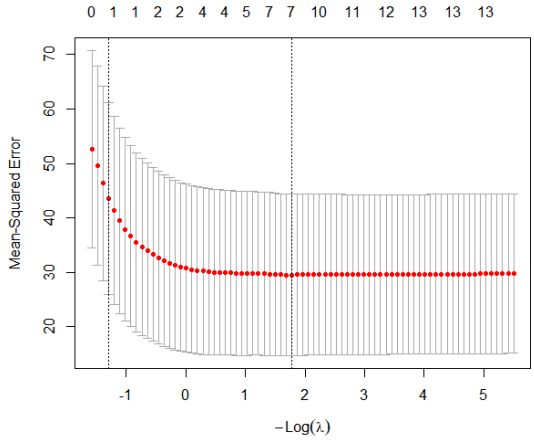
a) *Try out some of the regression methods explored in this chapter, such as best subset selection, the lasso, ridge regression, and PCR. Present and discuss results for the approaches that you consider.*

(code)

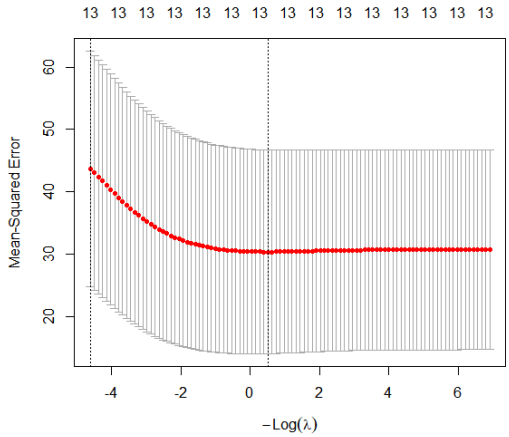
```
library(MASS)
library(glmnet)

data(Boston)
set.seed(333)

n<-nrow(Boston)
test_indices<-sample(n,size=floor(0.2*n))
train_data<-Boston[-test_indices,]
test_data<-Boston[test_indices,]

#stepwise
null_model<-lm(crim~1,data=train_data)
full_model<-lm(crim~.,data = train_data)
stepwise_model<-step(full_model,direction="both",scope=list(lower=null_model,
upper=full_model))

pred_stepwise<-predict(stepwise_model, newdata=test_data)
rmse_stepwise<-sqrt(mean((test_data$crim-pred_stepwise)^2))
cat("Stepwise RMSE:", rmse_stepwise, "\n")
```

Stepwise RMSE: 9.937905

```
#Ridge&LASSO
x_train<-model.matrix(crim~., data=train_data)[,-1]
y_train<-train_data$crim
x_test<-model.matrix(crim~., data=test_data)[,-1]
y_test<-test_data$crim

ridge_cv<-cv.glmnet(x_train, y_train, alpha=0, nfolds=10,lambda=10^(seq(-3,2,length.out=100)),
standardize=TRUE)
lasso_cv<-cv.glmnet(x_train, y_train, alpha=1, nfolds=10, standardize=TRUE)
plot(lasso_cv)
```




```
plot(ridge_cv)
```




```
lambda_min_ridge<-ridge_cv$lambda.min
lambda_1se_ridge<-ridge_cv$lambda.1se
lambda_min_lasso<-lasso_cv$lambda.min
lambda_1se_lasso<-lasso_cv$lambda.1se

pred_lasso_1se<-predict(lasso_cv,newx= x_test, s=lambda_1se_lasso)
pred_ridge_1se<-predict(ridge_cv, newx = x_test, s=lambda_1se_ridge)

rmse_lasso<-sqrt(mean((y_test-pred_lasso_1se)^2))
rmse_ridge<-sqrt(mean((y_test-pred_ridge_1se)^2))

cat("LASSO RMSE:",rmse_lasso, "\n" )
cat("Ridge RMSE:",rmse_ridge, "\n" )
```

LASSO RMSE: 11.87896
Ridge RMSE: 11.80105

(comment)

I evaluated Stepwise selection, LASSO, and Ridge regression on the Boston data set.
The Stepwise model yielded the lowest test RMSE (9.94), followed by Ridge (11.80) and LASSO (11.88). This suggests that Stepwise had the best out-of-sample performance among the three.

b) Propose a model (or set of models) that seem to perform well on this data set, and justify your answer. Make sure that you are evaluating model performance using validation set error, crossvalidation, or some other reasonable alternative, as opposed to using training error.

(comment)

Based on the test set RMSE, the Stepwise model appears to perform best and is therefore recommended. It showed the lowest prediction error on the validation set, which is a reliable measure of generalization. Although LASSO and Ridge provide regularization and may help in preventing overfitting in high-dimensional settings, they did not outperform Stepwise in this case.

c) Does your chosen model involve all of the features in the data set? Why or why not?

(comment)

No, the Stepwise model does not use all features in the dataset. It selects a subset of predictors based on their contribution to the model fit. This helps improve performance by reducing overfitting and enhancing model interpretability.

4. Logistic regression: Chapter 4: 6, 9, 13 a-d

6-a) Suppose we collect data for a group of students in a statistics class with variables X1 = hours studied, X2 = undergrad GPA, and Y =receive an A. We ft a logistic regression and produce estimated coeffcient, $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.

- Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.

(code)

```
beta_0<--6
beta_1<-0.05
beta_2<-1
X1<-40
X2<-3.5
log_odds<-beta_0+beta_1*X1+beta_2*X2
prob<-1/(1+exp(-log_odds))
print(prob)
```

0.3775407

(comment)

Estimated probability of receiving an A is 0.378 (37.8%)

6-b) How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?

(code)

```
target_prob<-0.5
target_hours<--(beta_0+beta_2*X2)/beta_1
print(target_hours)
```

50

(comment)

Student needs to study 50 hours to get 50% chance of receiving an A grade.

9-a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

(code)

```
odds<-0.37
p<-odds/(1+odds)
cat("Probability of default:", round(p,4))
```

(comment)

On average, 27% of people with 0.37 odds with default

9-b) Suppose that an individual has a 16 % chance of defaulting on her credit card payment. What are the odds that she will default?

(code)

```
p<-0.16
odds<-p/(1-p)
cat("Odds of default:", round(odds,4))
```

(comment)

Person who has 16% chance of default has 0.19 odds

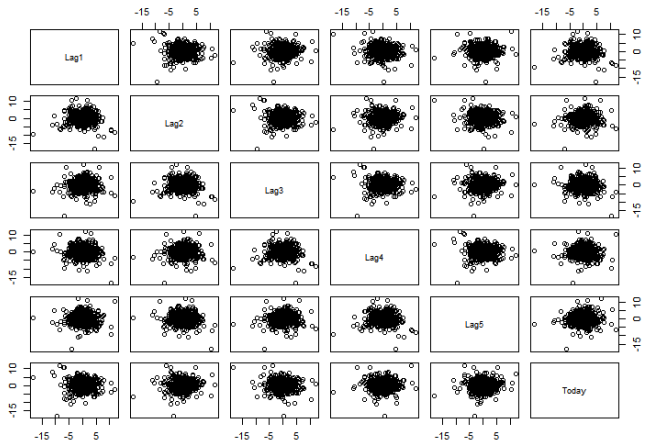
13-a) Produce some numerical and graphical summaries of the Weekly data. Do there appear to be any patterns?

(code)

```
data("Weekly")
str(Weekly)
summary(Weekly)
cor(Weekly[,sapply(Weekly, is.numeric)])
pairs(Weekly[,c("Lag1","Lag2","Lag3","Lag4","Lag5","Today")])
```

```
> summary(Weekly)
      Year           Lag1               Lag2               Lag3               Lag4               Lag5              Volume            Today          Direction
 Min.   :1990   Min.   :-18.1950   Min.   :-18.1950   Min.   :-18.1950   Min.   :-18.1950   Min.   :-18.1950   Min.   :0.08747   Min.   :-18.1950   Down:484
 1st Qu.:1995   1st Qu.: -1.1540   1st Qu.: -1.1540   1st Qu.: -1.1580   1st Qu.: -1.1580   1st Qu.: -1.1660   1st Qu.:0.33202   1st Qu.: -1.1540   Up  :605
 Median :2000   Median :  0.2410   Median :  0.2410   Median :  0.2410   Median :  0.2380   Median :  0.2340   Median :1.00268   Median :  0.2410
 Mean   :2000   Mean   :  0.1506   Mean   :  0.1511   Mean   :  0.1472   Mean   :  0.1458   Mean   :  0.1399   Mean   :1.57462   Mean   :  0.1499
 3rd Qu.:2005   3rd Qu.:  1.4050   3rd Qu.:  1.4090   3rd Qu.:  1.4090   3rd Qu.:  1.4090   3rd Qu.:  1.4050   3rd Qu.:2.05373   3rd Qu.:  1.4050
 Max.   :2010   Max.   : 12.0260   Max.   : 12.0260   Max.   : 12.0260   Max.   : 12.0260   Max.   : 12.0260
```

```
>   cor(Weekly[,sapply(weekly, is.numeric)])
              Year         Lag1        Lag2        Lag3
Year    1.00000000 -0.032289274 -0.03339001 -0.03000649
Lag1   -0.03228927  1.000000000 -0.07485305  0.05863568
Lag2   -0.03339001 -0.074853051  1.00000000 -0.07572091
Lag3   -0.03000649  0.058635682 -0.07572091  1.00000000
Lag4   -0.03112792 -0.071273876  0.05838153 -0.07539587
Lag5   -0.03051910 -0.008183096 -0.07249948  0.06065717
Volume  0.84194162 -0.064951313 -0.08551314 -0.06928771
Today  -0.03245989 -0.075031842  0.05916672 -0.07124364
               Lag4         Lag5      Volume        Today
Year    -0.031127923 -0.030519101  0.84194162 -0.032459894
Lag1    -0.071273876 -0.008183096 -0.06495131 -0.075031842
Lag2     0.058381535 -0.072499482 -0.08551314  0.059166717
Lag3    -0.075395865  0.060657175 -0.06928771 -0.071243639
Lag4     1.000000000 -0.075675027 -0.06107462 -0.007825873
Lag5    -0.075675027  1.000000000 -0.05851741  0.011012698
Volume  -0.061074617 -0.058517414  1.00000000 -0.033077783
Today   -0.007825873  0.011012698 -0.03307778  1.000000000
```



(comment)

Based on the numerical and graphical summaries of the Weekly dataset, there appear to be very weak or no clear patterns between past returns and the current week's return.

The pairwise scatterplots show no obvious linear relationships, and the correlation coefficients between these variables are all close to zero.

13-b) Use the full data set to perform a logistic regression with Direction as the response and the five lag variables plus Volume as predictors. Use the summary function to print the results. Do any of the predictors appear to be statistically significant? If so, which ones?

(code)

```
multi_logit<-glm(Direction~Lag1+Lag2+Lag3+Lag4+Lag5+Volume, data=Weekly, family=binomial)
summary(multi_logit)
```

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.26686    0.08593   3.106   0.0019 **
Lag1        -0.04127    0.02641  -1.563   0.1181
Lag2         0.05844    0.02686   2.175   0.0296 *
Lag3        -0.01606    0.02666  -0.602   0.5469
Lag4        -0.02779    0.02646  -1.050   0.2937
Lag5        -0.01447    0.02638  -0.549   0.5833
Volume      -0.02274    0.03690  -0.616   0.5377
```

(comment)

Among the predictors, only Lag2 is statistically significant ($p = 0.0296$), while the others do not show a significant relationship with the response variable.

13-c) Compute the confusion matrix and overall fraction of correct predictions. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.

(code)

```
Weekly$prob_multi<-predict(multi_logit,type="response")
Weekly$pred_05<-ifelse(Weekly$prob_multi>0.5, "Up", "Down")
Weekly$pred_05<-factor(Weekly$pred_05, levels=c("Down","Up"))
Weekly$Direction<-factor(Weekly$Direction, levels=c("Down", "Up"))
```

```
conf_05<- table(Predicted=Weekly$pred_05, Actual=Weekly$Direction)
print("Confusion Matrix at 0.5 threshold:")
print(conf_05)
accuracy<-(554+57)/(54+48+430+557)
print(accuracy)
```

```
         Actual
Predicted Down  Up
     Down   54  48
     Up    430 557
>   accuracy<-(554+57)/(54+48+430+557)
>   print(accuracy)
[1] 0.5610652
```

(comment)

The confusion matrix shows the model predicts "Up" much more often than "Down".

It correctly classified 611 out of 1089 observations, giving an accuracy of 56.1%. This suggests the model tends to overpredict "Up" and struggles with identifying "Down" weeks.

13-d) Now ft the logistic regression model using a training data period from 1990 to 2008, with Lag2 as the only predictor. Compute the confusion matrix and the overall fraction of correct predictions for the held out data (that is, the data from 2009 and 2010).

(code)

```
train_data<-Weekly[Weekly$Year <= 2008,]
test_data<-Weekly[Weekly$Year>2008,]

logit_model<-glm(Direction~Lag2, data=train_data, family=binomial)
pred_probs<-predict(logit_model, newdata=test_data, type="response")

pred_class<-ifelse(pred_probs>0.5,"Up","Down")
pred_class<-factor(pred_class,levels=c("Down","Up"))

actual_class<-test_data$Direction
conf_matrix<-table(Predicted=pred_class, Actual=actual_class)
print(conf_matrix)
```

(comment)

I trained a logistic regression model using data from 1990 to 2008 with Lag2 as the only predictor. Using this model on the 2009–2010 test data, we obtained a confusion matrix with 56 true positives and 9 true negatives. The overall accuracy is approximately 62.5%, suggesting moderate predictive performance on the held-out data.

5. Regression Trees: Chapter 8: #8 BUT: Use the Austin housing data from the prediction contest (austinhousing.csv) instead of the dataset in the book. Use the following variables to generate predictions for log(latestPrice): latitude, longitude, hasAssociation, livingAreaSqFt, numOfBathrooms, numOfBedrooms. (See the description of the dataset in the individual prediciton project assignment.) When reporting your prediction errors, report them in terms of prices (not log prices).

a) Split the data set into a training set and a test set.

(code)

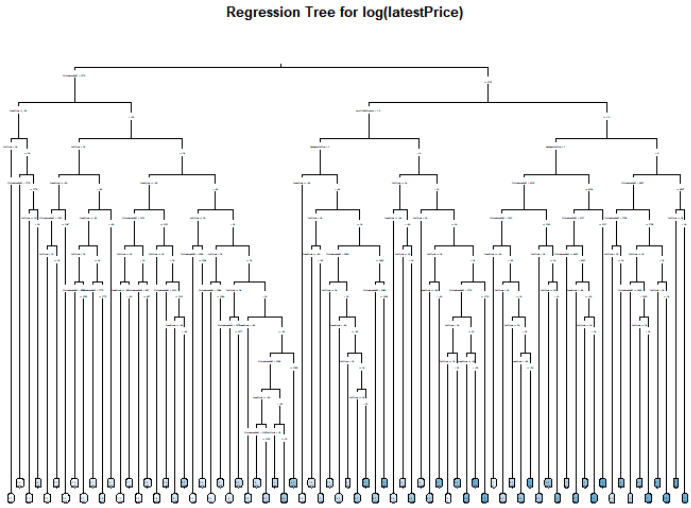
```
library(tidyverse)

data<-read.csv("austinhouses.csv")
data<-data %>%
select(latestPrice,latitude, longitude, hasAssociation, livingAreaSqFt,
      numOfBathrooms, numOfBedrooms) %>%
      drop_na()

data$logPrice<-log(data$latestPrice)

set.seed(111)
n<-nrow(data)
```

```
train_index<-sample(1:n, size=0.8*n)
train_data<-data[train_index,]
test_data<-data[-train_index,]
```

b) Fit a regression tree to the training set. Plot the tree, and interpret the results. What test MSE do you obtain?

(code)

```
library(rpart)
library(rpart.plot)

tree_model <- rpart(logPrice ~ latitude + longitude + hasAssociation +
                    livingAreaSqFt + numOfBathrooms + numOfBedrooms,
                      data = train_data,
                      method = "anova",
                      control = rpart.control(cp = 0.001))

rpart.plot(tree_model, type = 3, extra = 1, fallen.leaves = TRUE,
           main = "Regression Tree for log(latestPrice)")

log_preds <- predict(tree_model, newdata = test_data)
price_preds <- exp(log_preds)
test_mse <- mean((test_data$latestPrice - price_preds)^2)
cat("Test MSE (on original price scale):", round(test_mse, 2), "\n")
```

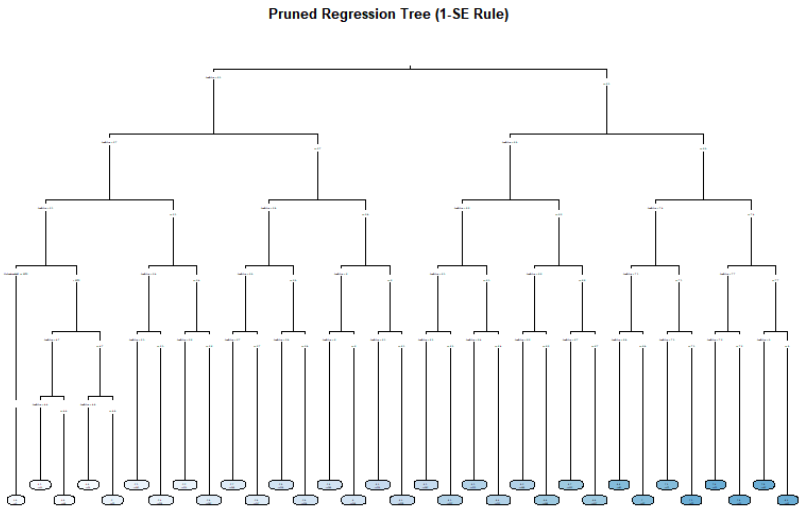


Test MSE (on original price scale): 30339.5

c) Use cross-validation in order to determine the optimal level of tree complexity. Does pruning the tree improve the test MSE?

(code)

```
tree_full <- rpart(log(latestPrice) ~ ., data = train_data, method = "anova",
                   control = rpart.control(cp = 0.0001, minsplit = 10))

plotcp(tree_full)
print(tree_full$cptable)

min_row <- which.min(tree_full$cptable[, "xerror"])
min_cp <- tree_full$cptable[min_row, "CP"]

se_threshold <- tree_full$cptable[min_row, "xerror"] + tree_full$cptable[min_row, "xstd"]
se_row <- min(which(tree_full$cptable[, "xerror"] <= se_threshold))
se_cp <- tree_full$cptable[se_row, "CP"]

tree_pruned <- prune(tree_full, cp = se_cp)

rpart.plot(tree_pruned, type = 3, extra = 1, fallen.leaves = TRUE,
           main = "Pruned Regression Tree (1-SE Rule)")

preds_pruned <- predict(tree_pruned, newdata = test_data)
mse_pruned <- mean((exp(preds_pruned) - test_data$latestPrice)^2)
cat("Test MSE (Pruned Tree):", round(mse_pruned, 2), "\n")

preds_full <- predict(tree_full, newdata = test_data)
mse_full <- mean((exp(preds_full) - test_data$latestPrice)^2)
cat("Test MSE (Full Tree):", round(mse_full, 2), "\n")
```



Test MSE (Pruned Tree): 505.71

Test MSE (Full Tree): 845.28

(comment)

Pruning based on the 1-SE rule helped reduce the test MSE from 845.28 to 505.71.

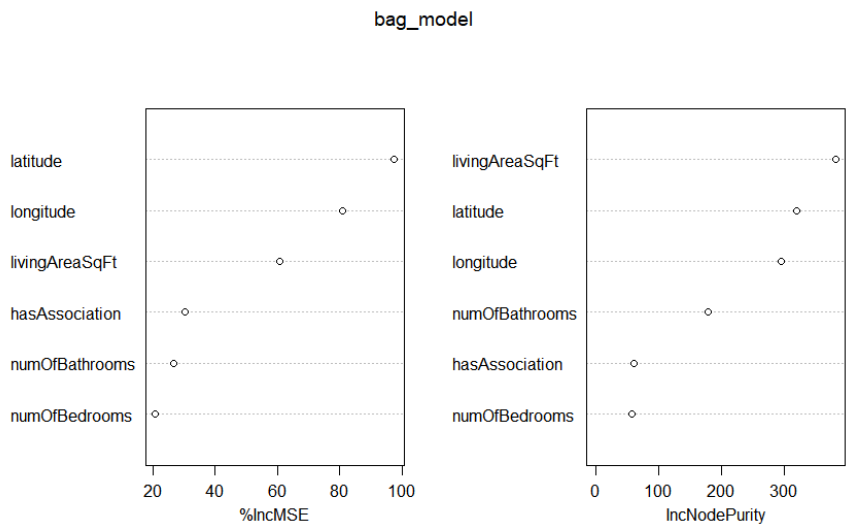
The full tree was too complex and overfit the training data, while the pruned tree performed better on the test set.

d) Use the bagging approach in order to analyze this data. What test MSE do you obtain? Use the importance() function to determine which variables are most important.

(code)

```
library(randomForest)

set.seed(111)
bag_model <- randomForest( logPrice ~ latitude + longitude + hasAssociation + livingAreaSqFt
                           + numOfBathrooms + numOfBedrooms, data = train_data,
                             importance = TRUE)

preds_bag_log <- predict(bag_model, newdata = test_data)
mse_bag <- mean((exp(preds_bag_log) - test_data$latestPrice)^2)

cat("Test MSE (Bagging, using formula):", round(mse_bag, 2), "\n")

importance(bag_model)
varImpPlot(bag_model)
```

Test MSE (Bagging, using formula): 22315.19



(comment)

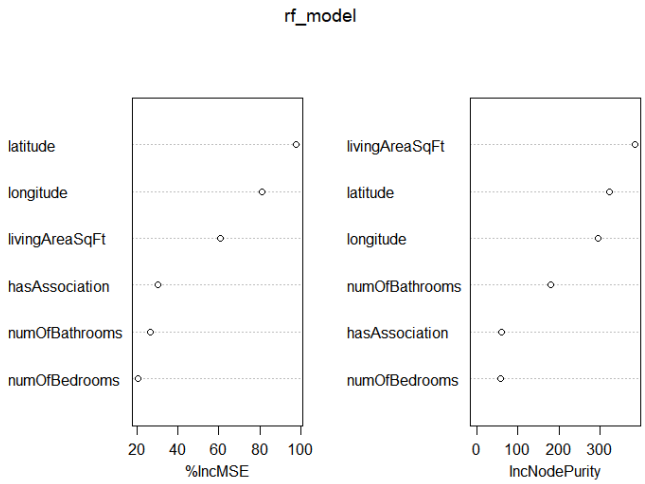
Based on the importance() function, the most critical variables are latitude and livingAreaSqFt.

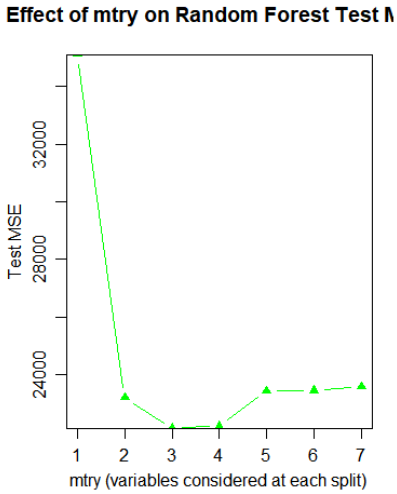
e) Use random forests to analyze this data. What test MSE do you obtain? Use the importance() function to determine which variables are most important. Describe the effect of m, the number of variables considered at each split, on the error rate obtained.

(code)

```
set.seed(111)
rf_model <- randomForest( logPrice ~ latitude + longitude + hasAssociation + livingAreaSqFt +
                          numOfBathrooms + numOfBedrooms, data = train_data,
                          importance = TRUE)

rf_preds_log <- predict(rf_model, newdata = test_data)
rf_preds_price <- exp(rf_preds_log)

mse_rf <- mean((rf_preds_price - test_data$latestPrice)^2)
cat("Test MSE (Random Forest):", round(mse_rf, 2), "\n")

importance(rf_model)
varImpPlot(rf_model)

set.seed(111)
p <- ncol(train_data) - 1
mtry_vals <- 1:p
mse_vals <- numeric(length(mtry_vals))

for (i in seq_along(mtry_vals)) {model <- randomForest( logPrice ~ latitude + longitude + hasAssociation
                                                         + livingAreaSqFt +
                                                           numOfBathrooms + numOfBedrooms,
                                                         data = train_data, mtry = mtry_vals[i])
  preds_log <- predict(model, newdata = test_data)
  preds_price <- exp(preds_log)
  mse_vals[i] <- mean((preds_price - test_data$latestPrice)^2)}

  plot(mtry_vals, mse_vals, type = "b", pch = 17, col = "green",
       xlab = "mtry (variables considered at each split)",
       ylab = "Test MSE",
       main = "Effect of mtry on Random Forest Test MSE")
```

Test MSE (Random Forest): 22315.19





(comment)

I used random forests to predict log(latestPrice), and the test MSE (converted back to original scale) was 22,315.19. According to importance(), the most important variable was livingAreaSqFt, followed by latitude and longitude. I also tuned mtry and found that the test error was lowest when mtry = 3. After that, the error slightly increased, suggesting that using too many variables per split may hurt performance.

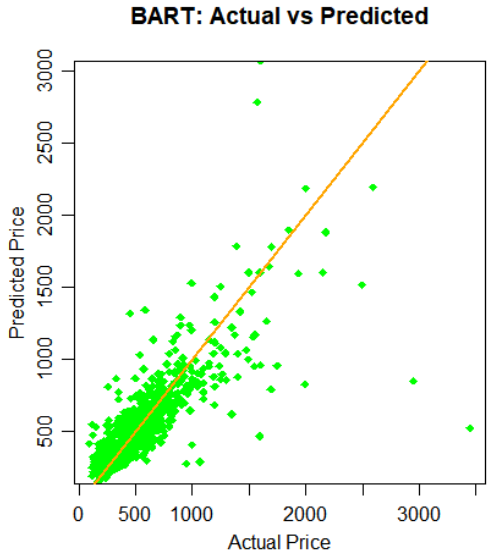
f) Now analyze the data using BART, and report your results.

(code)

```
library(BART)
x_train <- train_data %>%
   dplyr::select(latitude, longitude, hasAssociation, livingAreaSqFt,
                 numOfBathrooms, numOfBedrooms) %>%
   as.matrix()
y_train <- train_data$logPrice
x_test <- test_data %>%
   dplyr::select(latitude, longitude, hasAssociation, livingAreaSqFt,
                 numOfBathrooms, numOfBedrooms) %>%
   as.matrix()
y_test <- test_data$latestPrice

set.seed(111)
bart_model <- wbart(x.train = x_train, y.train = y_train,
                    x.test = x_test)
yhat_test_log <- apply(bart_model$yhat.test, 2, mean)

yhat_test <- exp(yhat_test_log)
mse_bart <- mean((yhat_test - y_test)^2)
cat("Test MSE (BART):", round(mse_bart, 2), "\n")

plot(y_test, yhat_test,
     xlab = "Actual Price", ylab = "Predicted Price",
```

```
        main = "BART: Actual vs Predicted", pch = 18, col = "green")
abline(0, 1, col = "orange", lwd = 2)
```

Test MSE (BART): 31165.16



(comment)

Using BART, the test MSE was 31,165.16. Most predictions aligned well with actual prices, especially in the lower to mid range. However, prediction errors tended to increase for higher-priced homes. Overall, BART performed reasonably well but did not outperform Random Forest.

6. Classification trees: Chapter 8: #11; in part c) use logistic regression.

a) Create a training set consisting of the frst 1,000 observations, and a test set consisting of the remaining observations.

(code)

```
library(ISLR2)
data(Caravan)
train_data<-Caravan[1:1000,]
test_data<-Caravan[-(1:1000),]
```

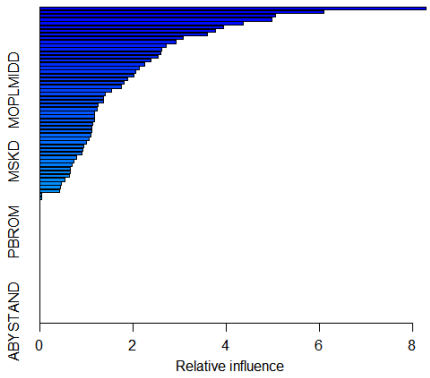
b) Fit a boosting model to the training set with Purchase as the response and the other variables as predictors. Use 1,000 trees, and a shrinkage value of 0.01. Which predictors appear to be the most important?

(code)

```
library(gbm)
train_data$PurchaseBinary <- ifelse(train_data$Purchase == "Yes", 1, 0)
test_data$PurchaseBinary    <- ifelse(test_data$Purchase == "Yes", 1, 0)

set.seed(111)
boost_model <- gbm(PurchaseBinary ~ . -Purchase, data = train_data, distribution = "bernoulli",
                   n.trees = 1000, shrinkage = 0.01, interaction.depth = 3)
probs <- predict(boost_model, newdata = test_data, n.trees = 1000, type = "response")
preds <- ifelse(probs > 0.5, 1, 0)
```

```
table(Predicted = preds, Actual = test_data$PurchaseBinary)
summary(boost_model)
```

```
         Actual
Predicted    0    1
        0 4511  280
        1   22    9
```



(comment)

I fitted a boosting model with 1,000 trees and shrinkage = 0.01. The most important predictors were MINKM30 and MSKD. But overall performance was weak — only 9 buyers were correctly identified.

b) Use the boosting model to predict the response on the test data. Predict that a person will make a purchase if the estimated probability of purchase is greater than 20 %. Form a confusion matrix. What fraction of the people predicted to make a purchase do in fact make one? How does this compare with the result obtained from applying logistic regression to this dataset?

(code)

```
set.seed(111)
boost_model <- gbm(PurchaseBinary ~ . -Purchase, data = train_data, distribution = "bernoulli",
                   n.trees = 1000, shrinkage = 0.01)
probs_boost <- predict(boost_model, newdata = test_data, n.trees = 1000, type = "response")
preds_boost <- ifelse(probs_boost > 0.2, 1, 0)

cm_boost <- table(Predicted = preds_boost, Actual = test_data$PurchaseBinary)
print(cm_boost)

precision_boost <- cm_boost["1", "1"] / sum(cm_boost["1", ])
cat("Boosting Precision:", round(precision_boost, 3), "\n")

logit_model <- glm(PurchaseBinary ~ . -Purchase, data = train_data, family = "binomial")
probs_logit <- predict(logit_model, newdata = test_data, type = "response")
preds_logit <- ifelse(probs_logit > 0.2, 1, 0)

cm_logit <- table(Predicted = preds_logit, Actual = test_data$PurchaseBinary)
print(cm_logit)
```

```
precision_logit <- cm_logit["1", "1"] / sum(cm_logit["1", ])
cat("Logistic Regression Precision:", round(precision_logit, 3), "\n")
```

```
                    Actual
         Predicted    0    1
                 0 4415  257
                 1  118   32
```

Boosting Precision: 0.213

```
                    Actual
         Predicted    0    1
                 0 4183  231
                 1  350   58
```

Logistic Regression Precision: 0.142

(comment)

I compared the predictive performance of a boosting model and logistic regression on the Caravan dataset, using a 20% probability threshold. Boosting achieved a higher precision (0.213) compared to logistic regression (0.142), meaning that among people predicted to purchase, a greater proportion actually did under boosting.